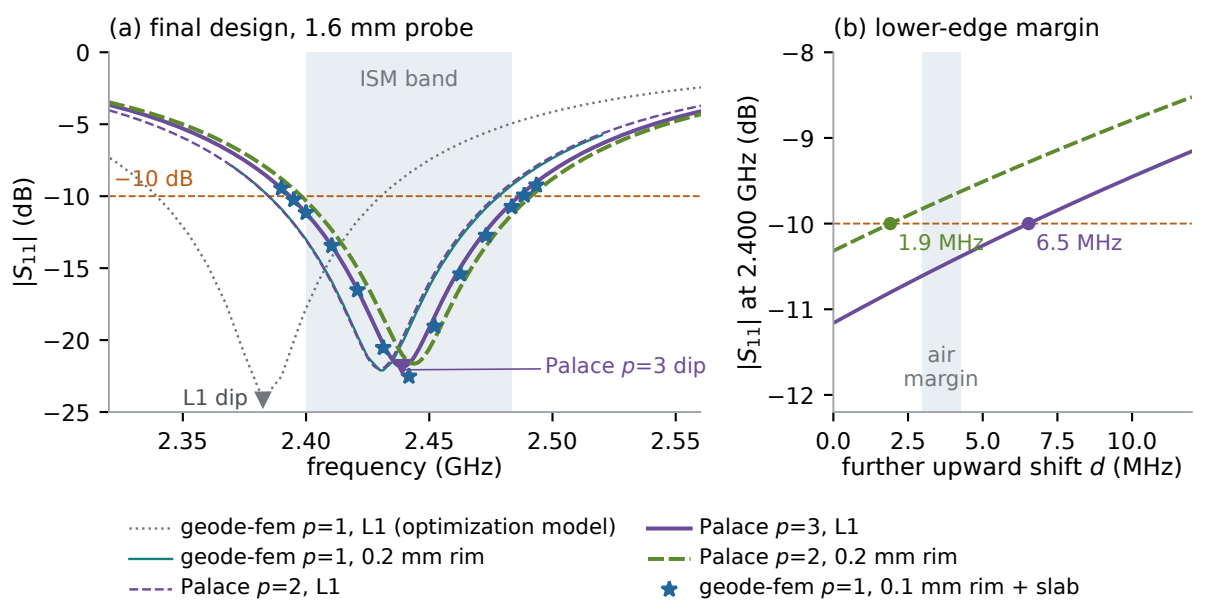
(a) final design, 1.6 mm probe
ISM band
−10 dB
L1 dip
Palace p=3 dip
|S11| (dB)
frequency (GHz)
(b) lower-edge margin
1.9 MHz
6.5 MHz
air
margin
|S11| at 2.400 GHz (dB)
further upward shift d (MHz)
geode-fem p=1, L1 (optimization model)
geode-fem p=1, 0.2 mm rim
Palace p=2, L1
Palace p=3, L1
Palace p=2, 0.2 mm rim
geode-fem p=1, 0.1 mm rim + slab